# AIAC Assignment - 15.5

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|---|---|---|

**Task 1 – Student Attendance API**

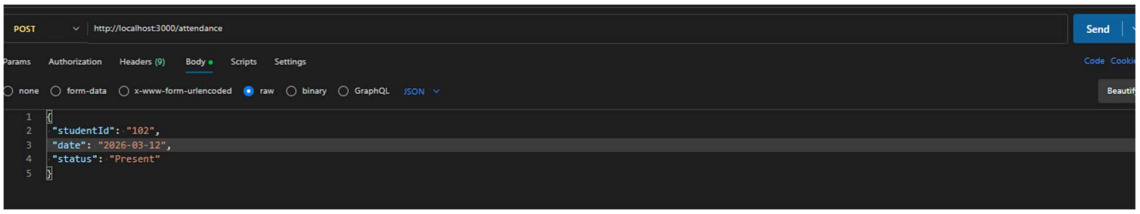
**Task:**
**Create a RESTful API for student attendance tracking.**

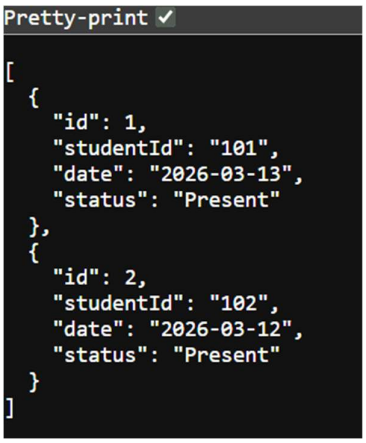
Instructions:
• Endpoints required:
o GET /attendance → View attendance records
o POST /attendance → Mark attendance
o PUT /attendance/{id} → Update attendance record
o DELETE /attendance/{id} → Remove attendance entry
• Ensure JSON-based request and response handling.

Expected Output:
• API for maintaining and updating attendance records.

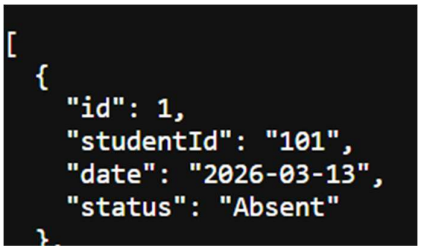
GET:

POST:

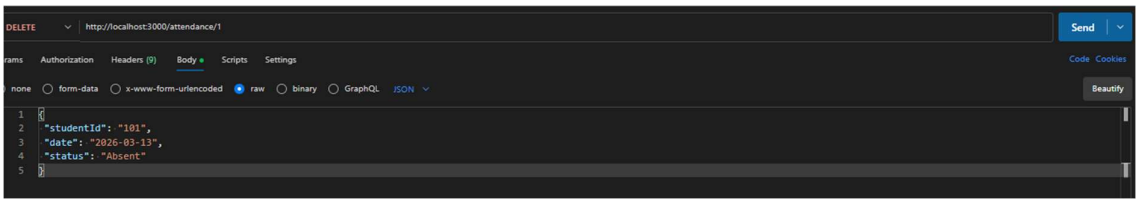


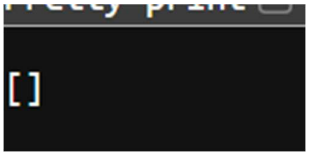



PUT:




DELETE:



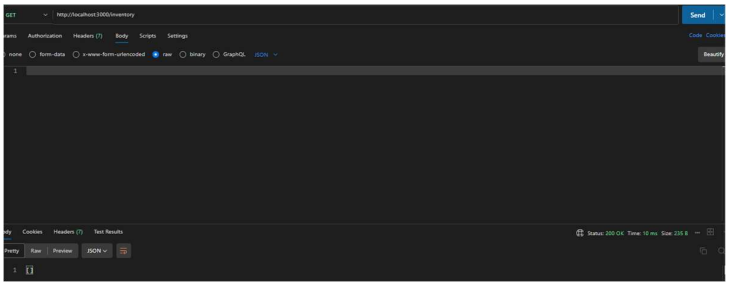

**Task 2 – Inventory Management API**

**Task:**
**Develop a REST API for warehouse inventory management.**

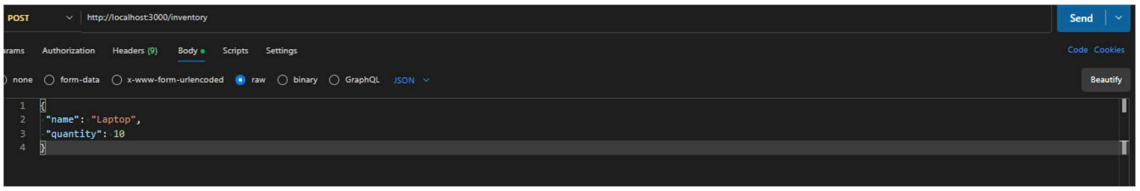
Instructions:
• Endpoints required:
o GET /inventory → List all items
o POST /inventory → Add new item
o PUT /inventory/{id} → Update stock quantity
o DELETE /inventory/{id} → Remove an item
• Implement validation for stock values.

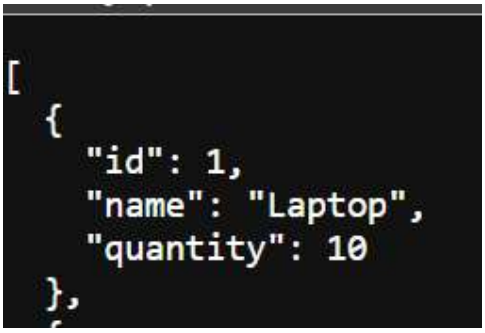
Expected Output:
• API capable of managing inventory stock levels.

GET:

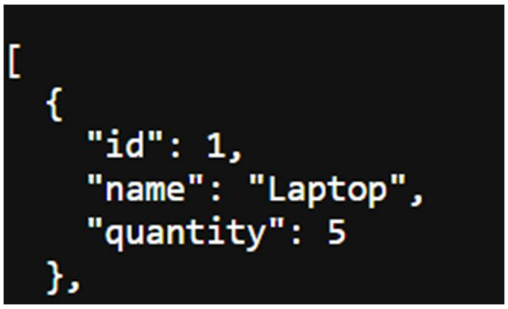


POST:

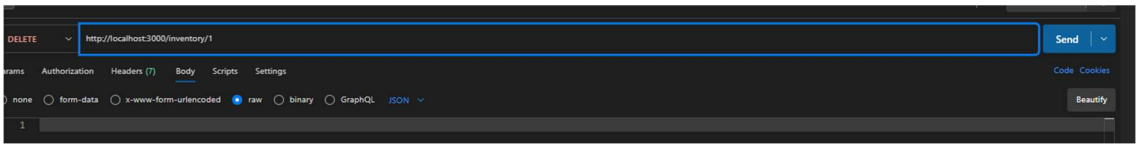


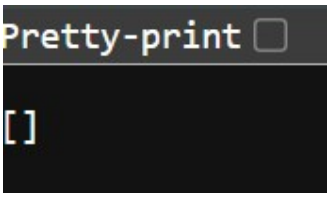




PUT:




DELETE:




**Task 3 – Hotel Room Booking API**

**Task:**
**Develop a RESTful API for hotel room booking.**

Instructions:

• Endpoints required:

o GET /rooms → View available rooms

o POST /rooms/book → Book a room

o GET /rooms/{id} → View booking details

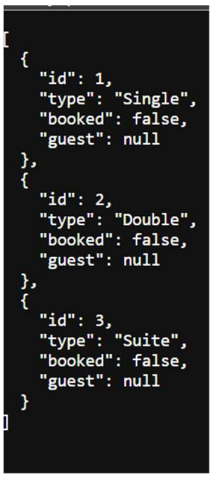
o DELETE /rooms/{id} → Cancel booking

• Implement error handling for unavailable rooms.

Expected Output:

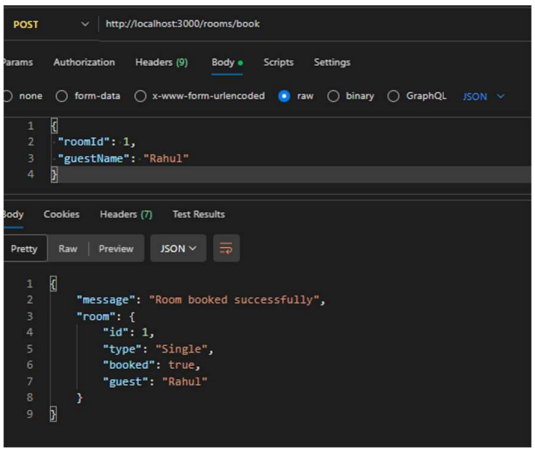
• API simulating hotel room booking operations.

GET:

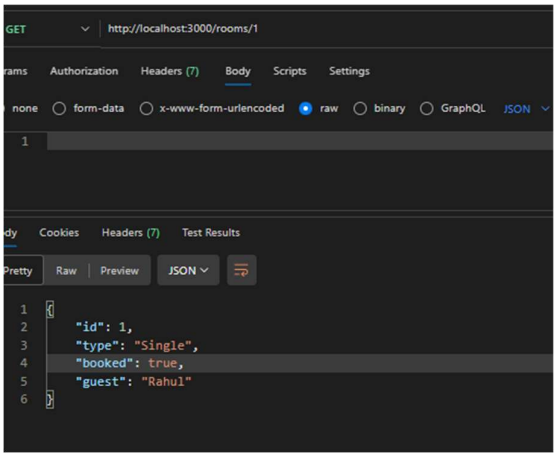



```
[
  {
    "id": 1,
    "type": "Single",
    "booked": false,
    "guest": null
  },
  {
    "id": 2,
    "type": "Double",
    "booked": false,
    "guest": null
  },
  {
    "id": 3,
    "type": "Suite",
    "booked": false,
    "guest": null
  }
]
```

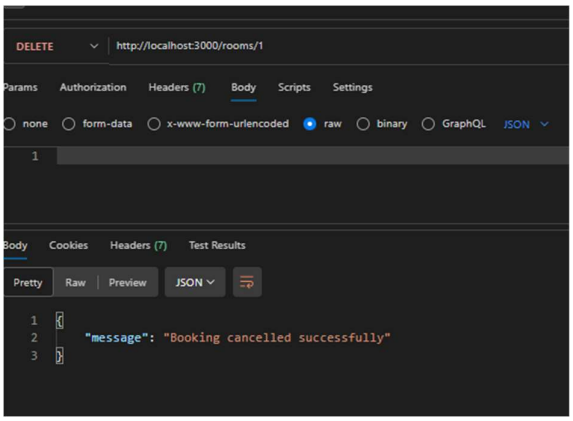

POST:



GET:




DELETE:



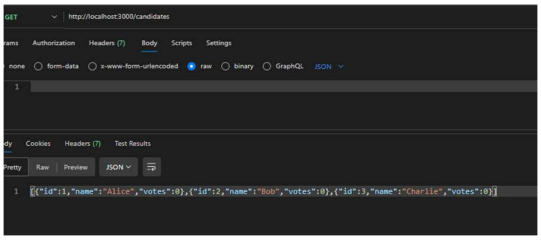

**Task 4 – Online Voting API**

**Task:**
**Create a RESTful API for an online voting system.**

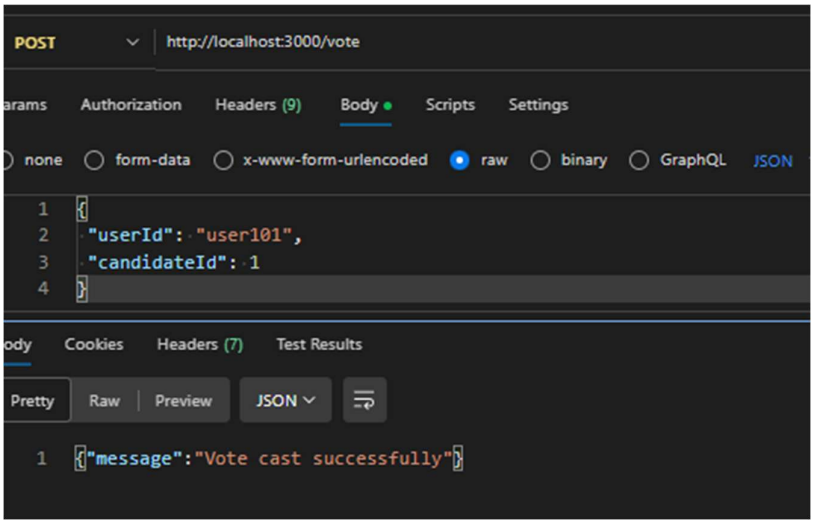
Instructions:
• Endpoints required:
o GET /candidates → List candidates
o POST /vote → Cast a vote
o GET /results → View voting results
• Ensure one vote per user using in-memory validation.

Expected Output:
• API simulating a secure online voting process.

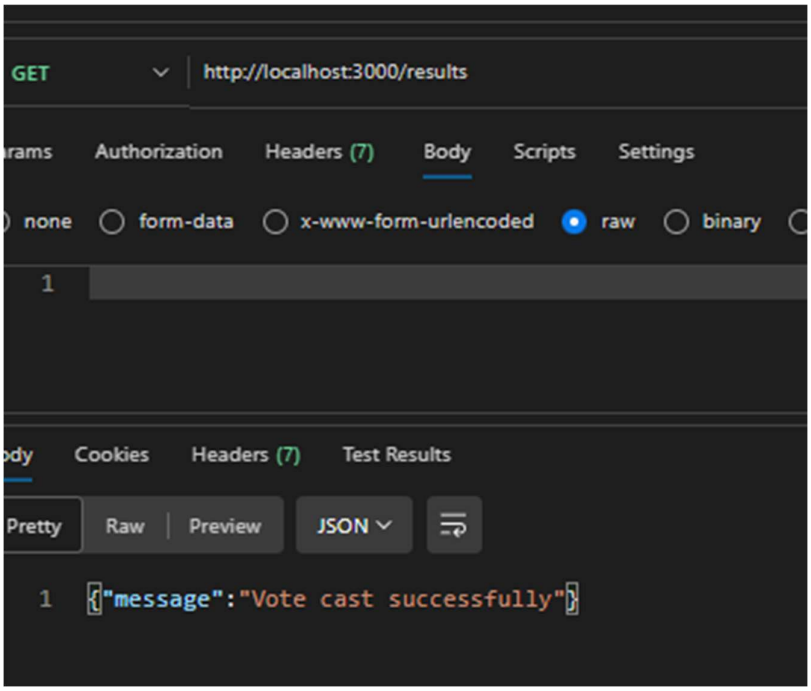
GET:



POST:




GET:




**Task 5 – Teacher Management API**

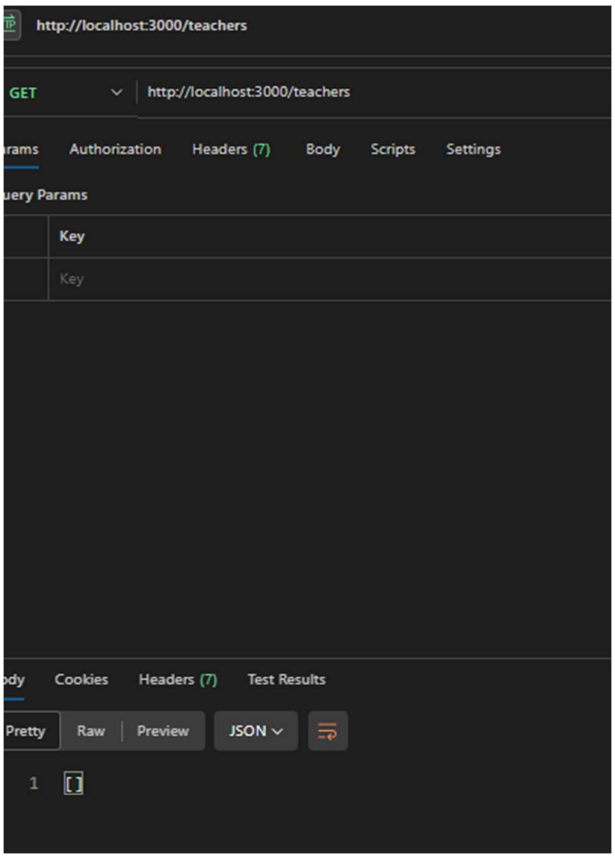
**Task:**
**Develop a RESTful API to manage teacher records.**

Instructions:
• Endpoints required:
o GET /teachers → List all teachers
o POST /teachers → Add a new teacher
o PUT /teachers/{id} → Update teacher details
o DELETE /teachers/{id} → Remove a teacher
• Ensure proper error handling and JSON responses.

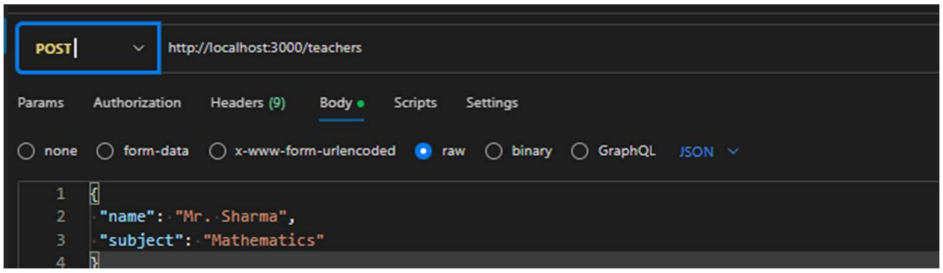
Expected Output:

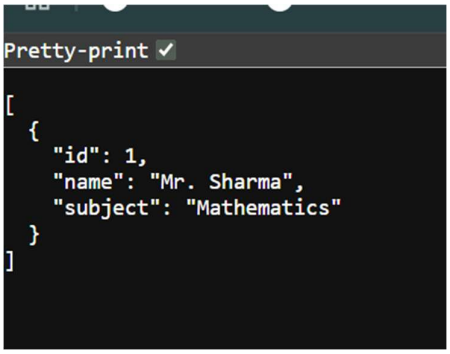
• Functional API for managing teacher information.

GET:

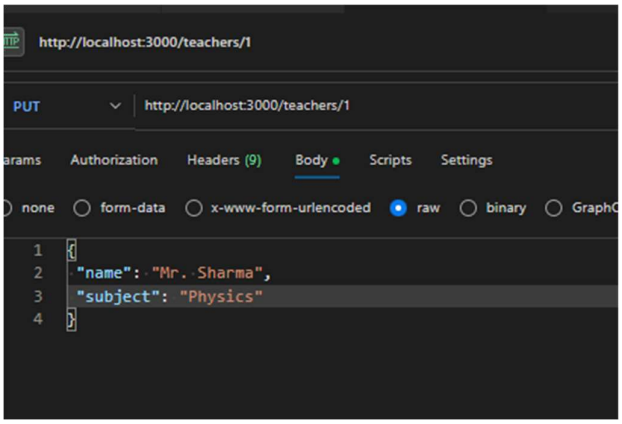


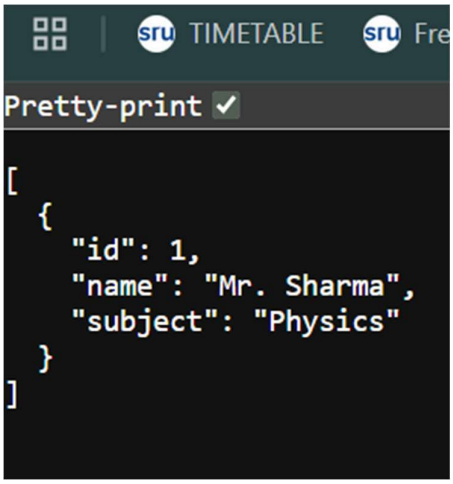

POST:



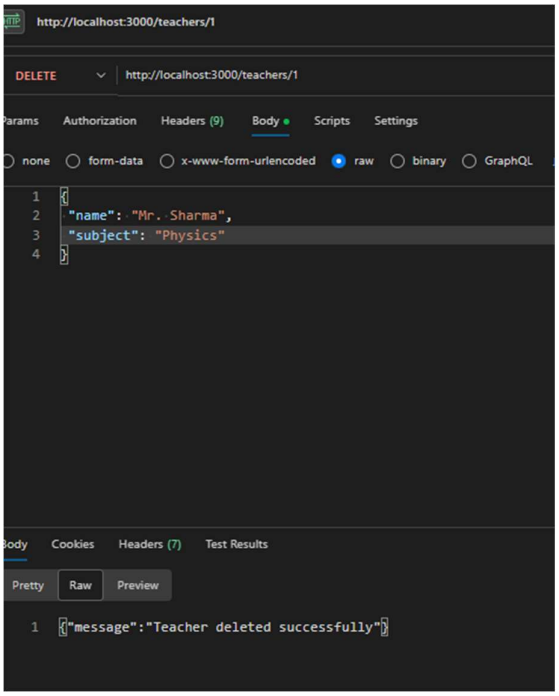



PUT:


http://localhost:3000/teachers/1
PUT
http://localhost:3000/teachers/1
Authorization
Headers (9)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
{
"name": "Mr. Sharma",
"subject": "Physics"
}


TIMETABLE
Pretty-print
[
{
"id": 1,
"name": "Mr. Sharma",
"subject": "Physics"
}
]

DELETE:


http://localhost:3000/teachers/1
DELETE
http://localhost:3000/teachers/1
Authorization
Headers (9)
Body
Scripts
Settings
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
{
"name": "Mr. Sharma",
"subject": "Physics"
}
Cookies
Headers (7)
Test Results
Pretty
Raw
Preview
{"message":"Teacher deleted successfully"}